

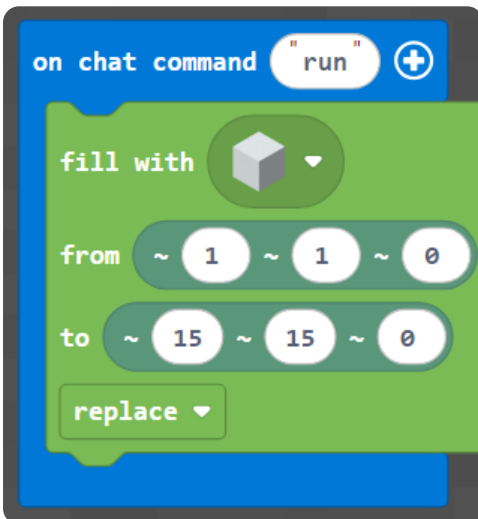
## 3D Coordinates · Pixel Art on the Wall (x, y) — English Worksheet (Intermediate)

**Topic:** Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate · **Time:** about 40 minutes

This lesson you copy a picture made of blocks onto your wall. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

**Build your canvas.** Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

### Blocks



### Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

### 1 · Predict 🧠

Read each set of steps. Write a sentence saying what you think you will see.

```
place red block at (5, 1)
place red block at (6, 1)
place red block at (7, 1)
```

**What do these three blocks make? Across or up? Which number stays the same?**

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```
place green block at (8, 13)
place green block at (8, 14)
place green block at (8, 15)
```

**These three share  $x = 8$ . Near the bottom or the top? Across or up?**

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```
place yellow block at (6, 1)
place yellow block at (7, 2)
place yellow block at (8, 3)
```

**Both numbers change each time. Does this make a straight row, a straight column, or a stair going up?**

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## 2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

**Bug A** — These should make the **bottom row of the pineapple**: five yellow blocks from (6, 1) to (10, 1). Right now they all land in the **same spot**.

```
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
```

**Hint:** to move across, the **x** has to change each time.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The apple stem should go **up** at x = 8, at (8, 13), (8, 14), (8, 15). Right now it grows **across** instead.

```
place brown block at (13, 8)
place brown block at (14, 8)
place brown block at (15, 8)
```

**Hint:** check which slot the changing number is in — x (across) or y (up). The two numbers got swapped.

**Write the fixed code:**

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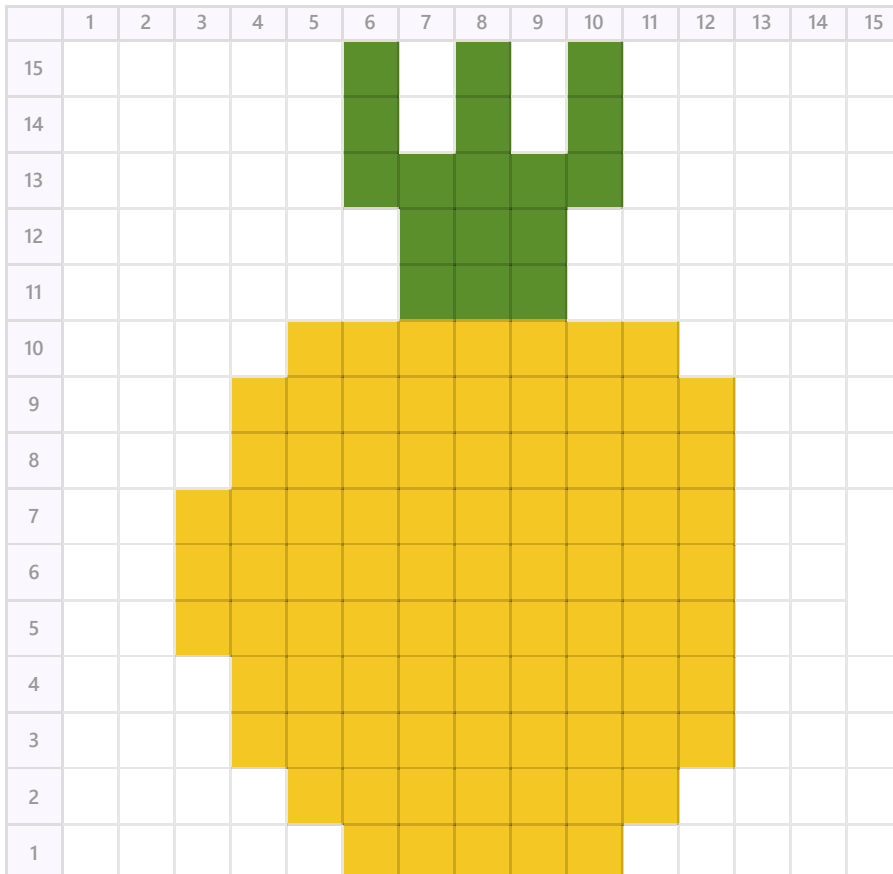
**Why was it wrong? Why does your fix work?**

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### 3 · Show Your Work 📷 🎥

Switch to your homework world and stand on your **home spot** (the red block). Copy the **pineapple** below onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

#### 🍷 Pineapple



Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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**Submit** 

Send this worksheet + your video to teacher on KakaoTalk.